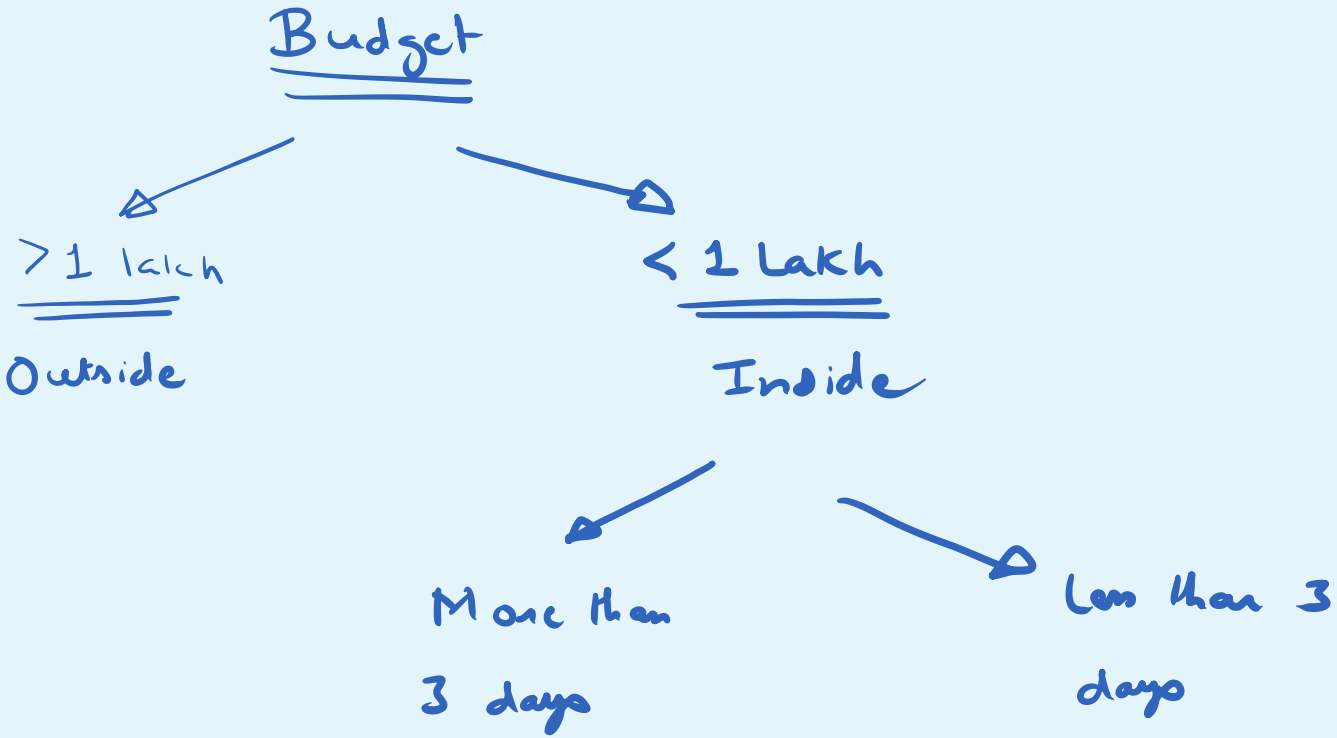
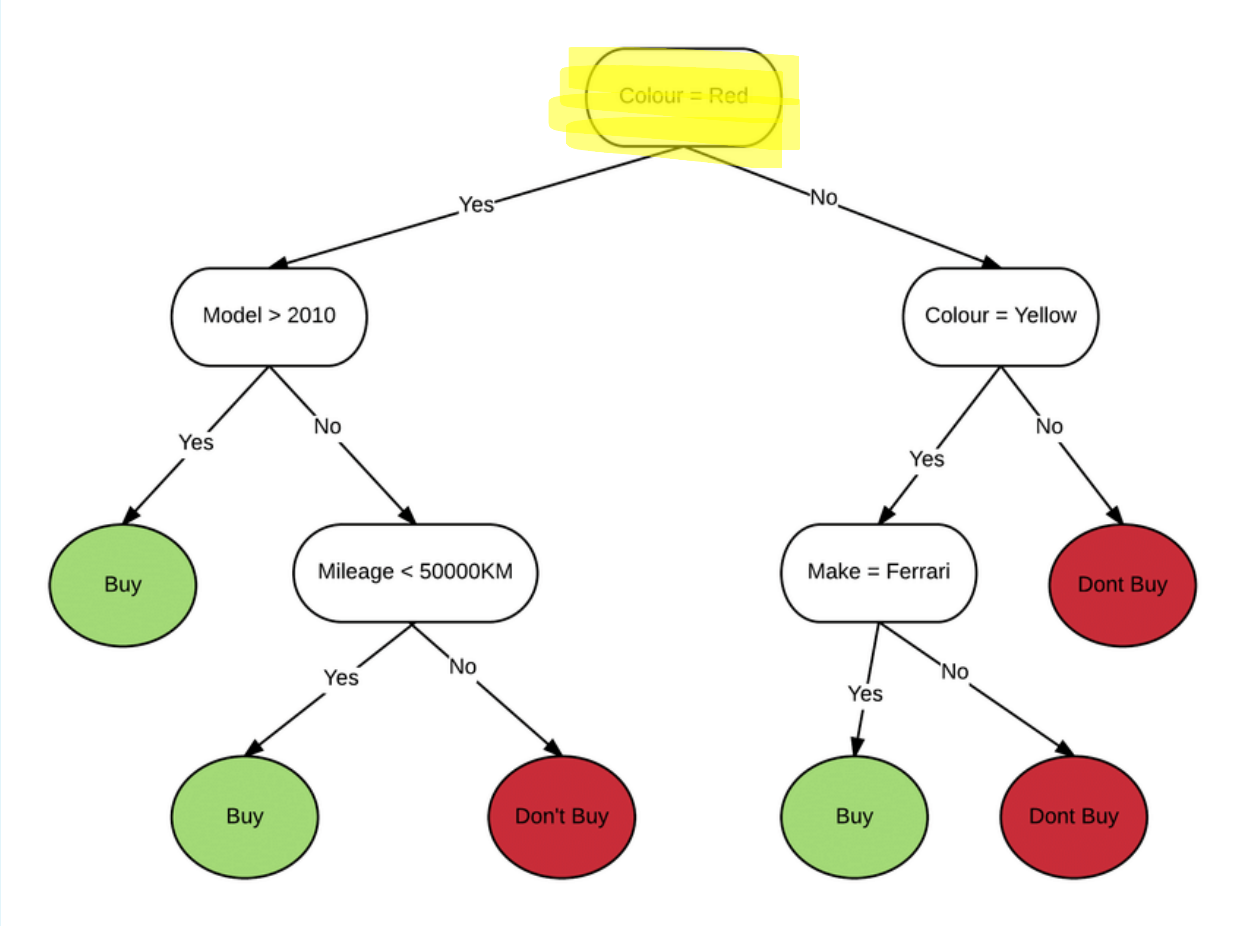
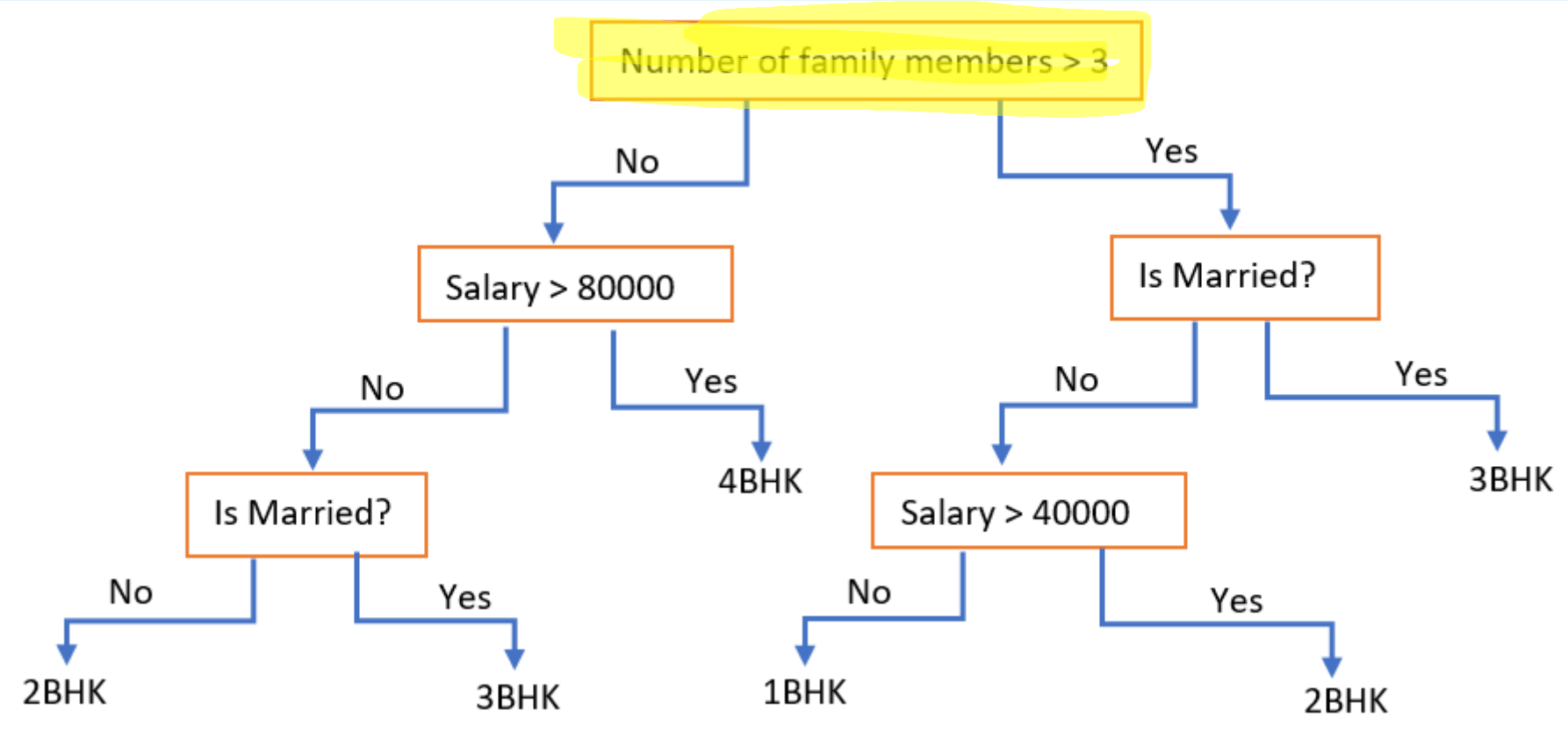
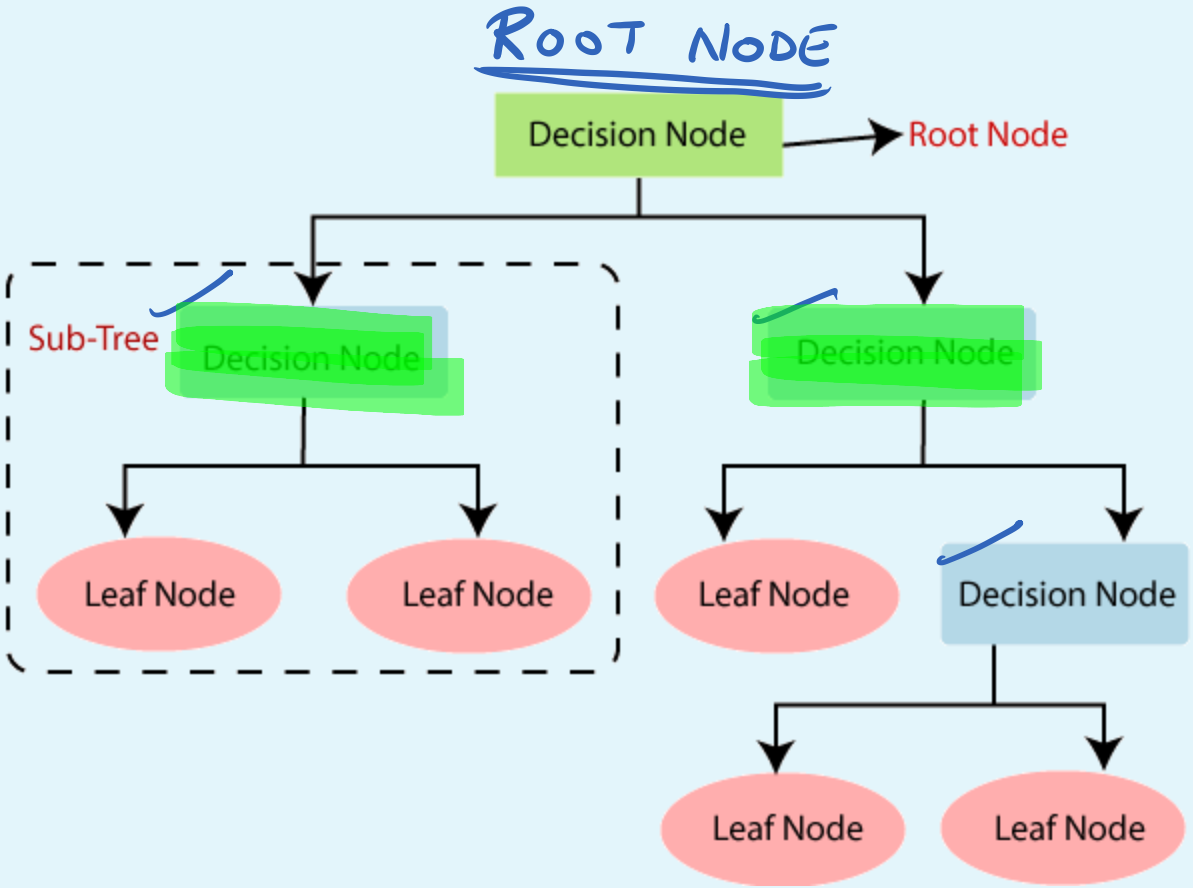
# Decision Trees

basic decision in life

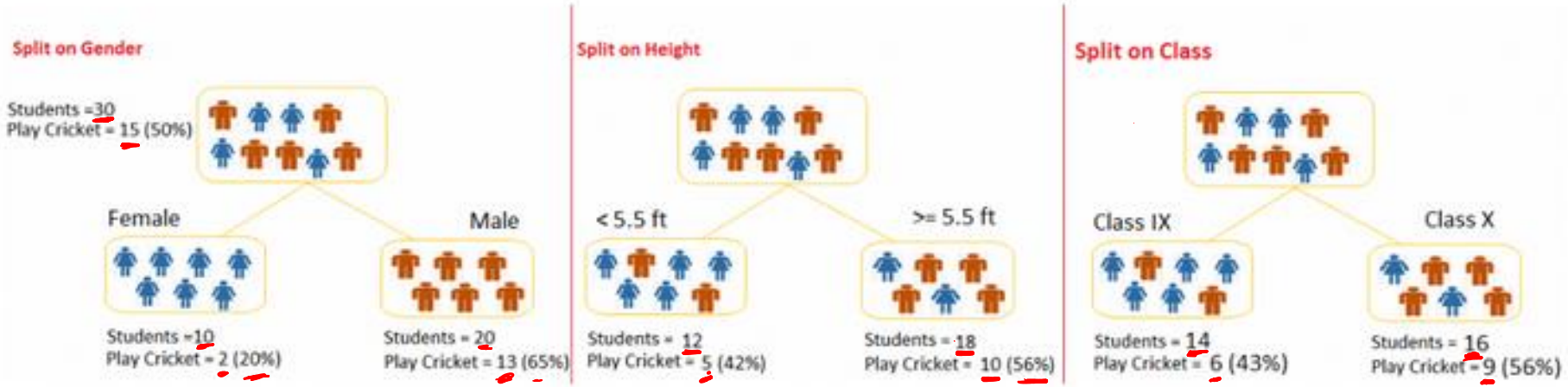



Colour = Red
Yes
No
Model > 2010
Colour = Yellow
Yes
No
Yes
No
Buy
Mileage < 50000KM
Make = Ferrari
Dont Buy
Yes
No
Yes
No
Buy
Don't Buy
Buy
Dont Buy


Number of family members > 3
No
Yes
Salary > 80000
Is Married?
No
Yes
No
Yes
4BHK
3BHK
Is Married?
Salary > 40000
No
Yes
No
Yes
2BHK
3BHK
1BHK
2BHK


ROOT NODE
Decision Node
Root Node
Sub-Tree
Decision Node
Decision Node
Leaf Node
Leaf Node
Leaf Node
Decision Node
Leaf Node
Leaf Node

Roots




inc asedcwantctocseparate--cplayingc ri ketcvscnotcplayingc ri ket

prob playing cricket

not prob playing cricket

Gini ⇒ Gender ⇒ Female $= (p)^2 + (1-p)^2$

$= (0.2)^2 + (0.8)^2 \Rightarrow 0.68$

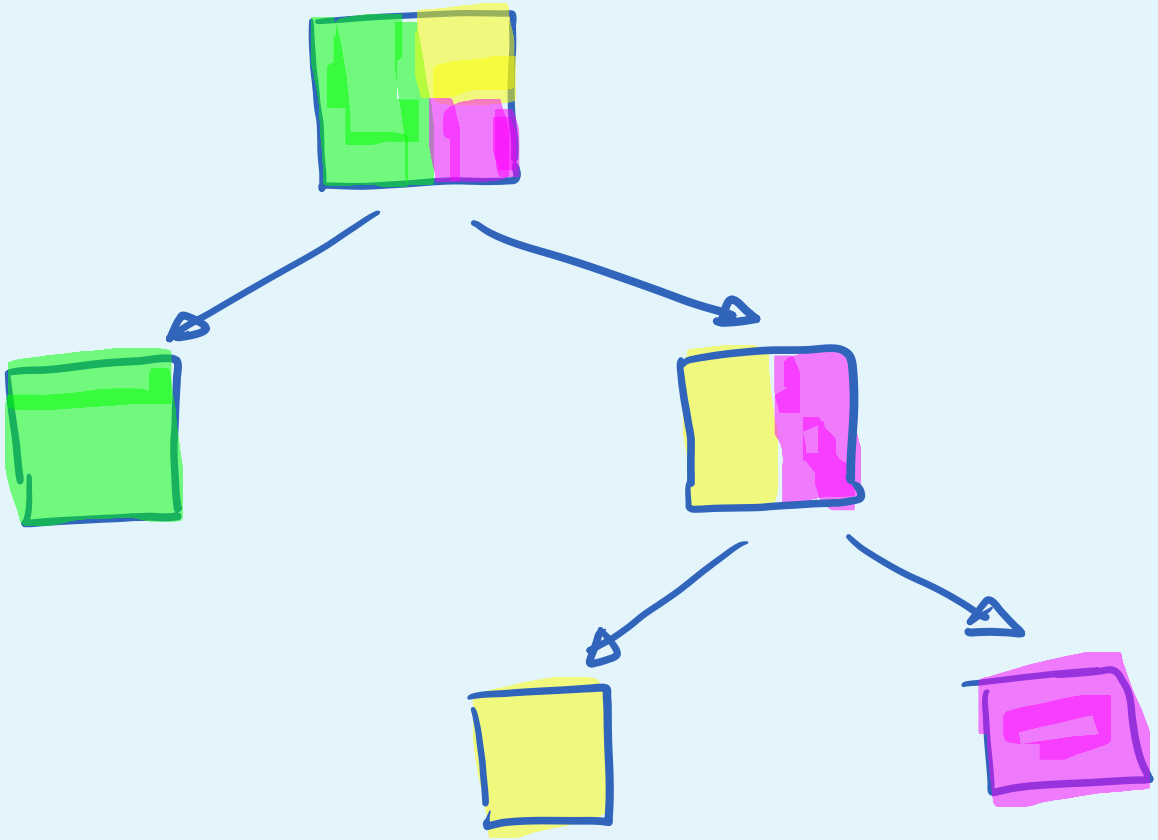
Male $= (0.65)^2 + (0.35)^2$

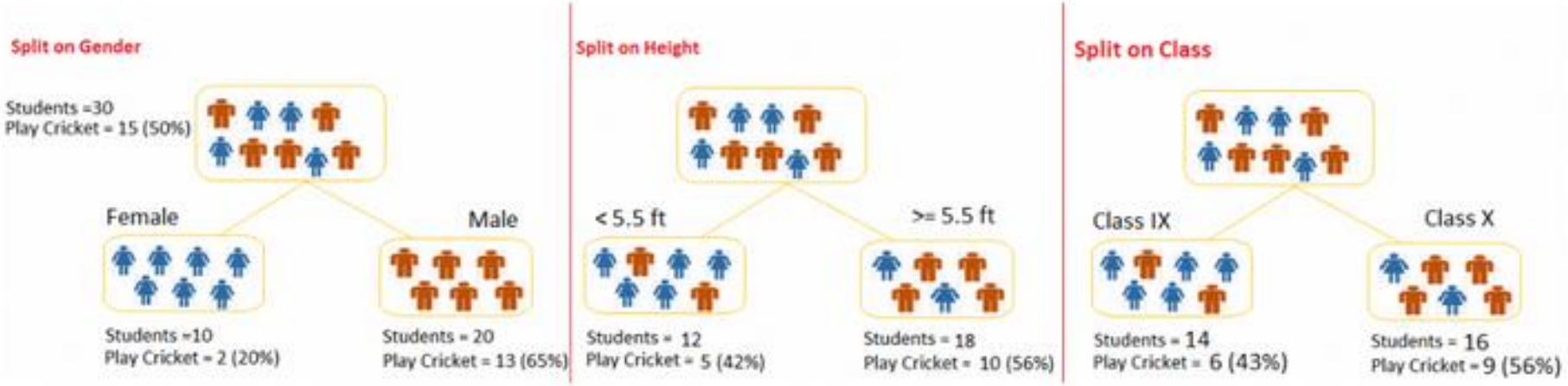
$= 0.42 + 0.12 \Rightarrow 0.54$

whenever we split data, we want to make sure we similar people in one group

on which variable you should make split

homogeneous example-- similar group






we cannot simply add these numbers so we will take weighted avg

Gini value (gender) $\Rightarrow \frac{10}{30} \times 0.68 + \frac{20}{30} \times 0.54$

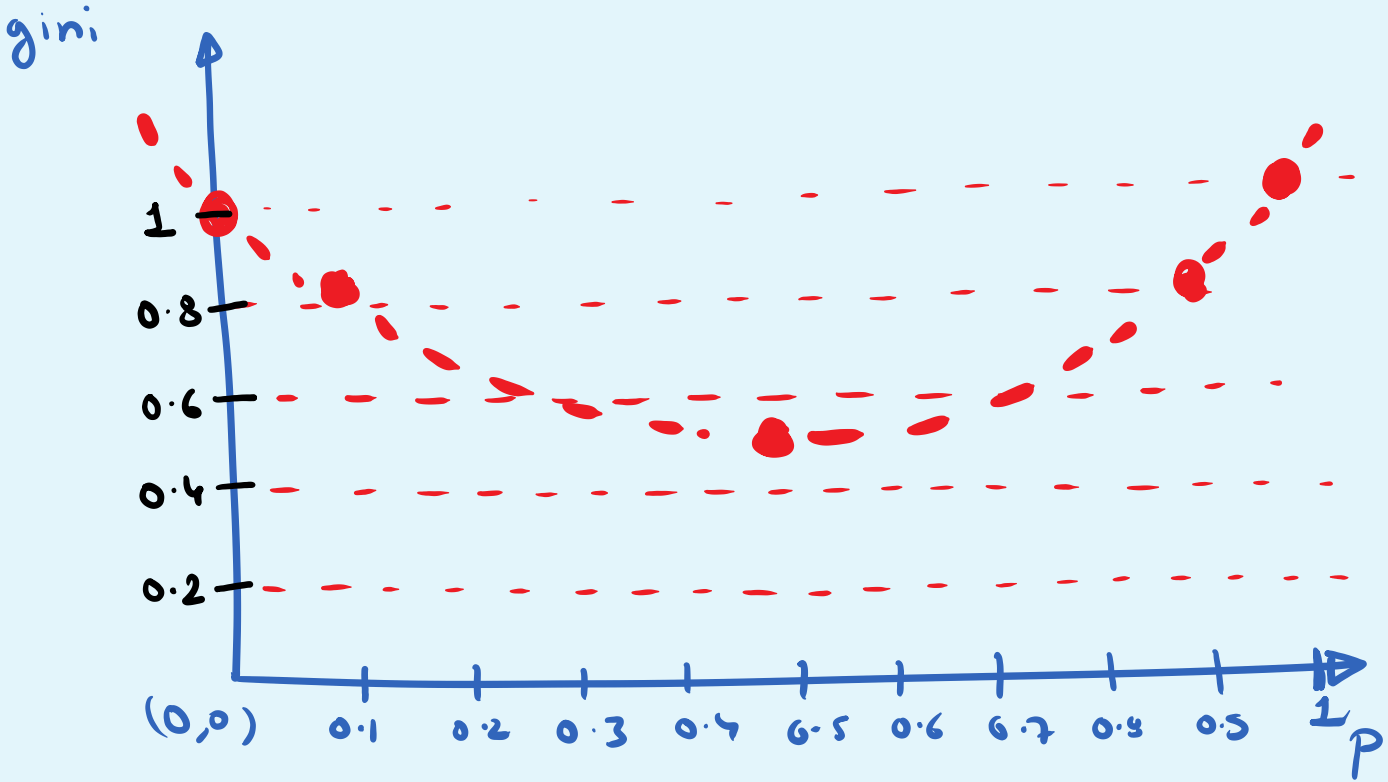
$\Rightarrow 0.226 + 0.36$

$\Rightarrow 0.586$

why are we taking p^2 and 1-p^2 we will understand by grapgh

range ?

$p = [0, 1]$




$$\text{Gini} = p^2 + (1-p)^2$$

**Ram** (ML)

1 Lakh / year

100 % accurate

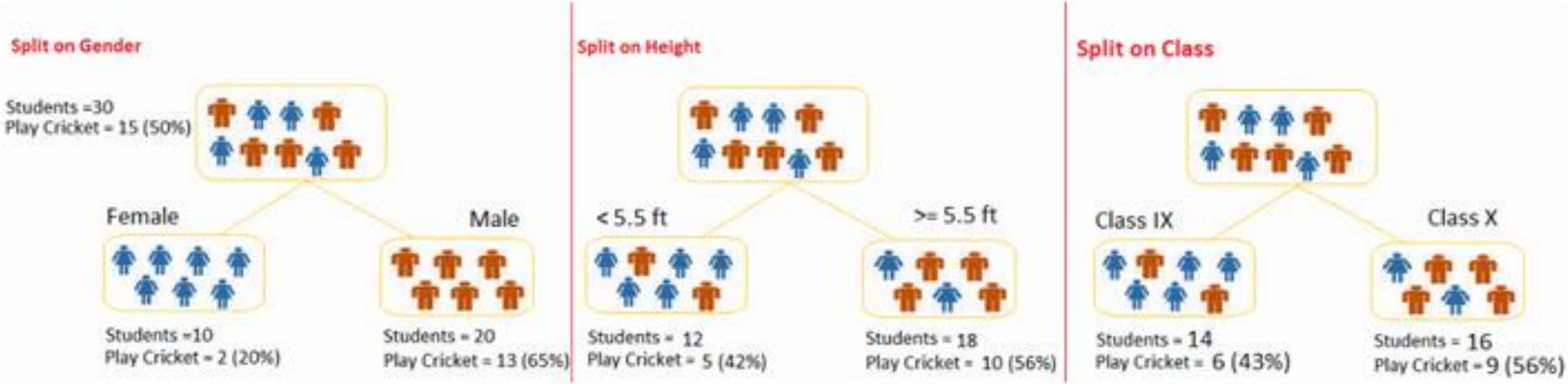
**Shyam** (ML)

10,000 / year

0 % accurate

ML

| | |
|---|---|
| 100% | 100% |
| 80% | 80% |
| 60% | 60% |
| 40% | 60% |
| 20% | 80% |
| 10% | 90% |
| 50% | 50% |

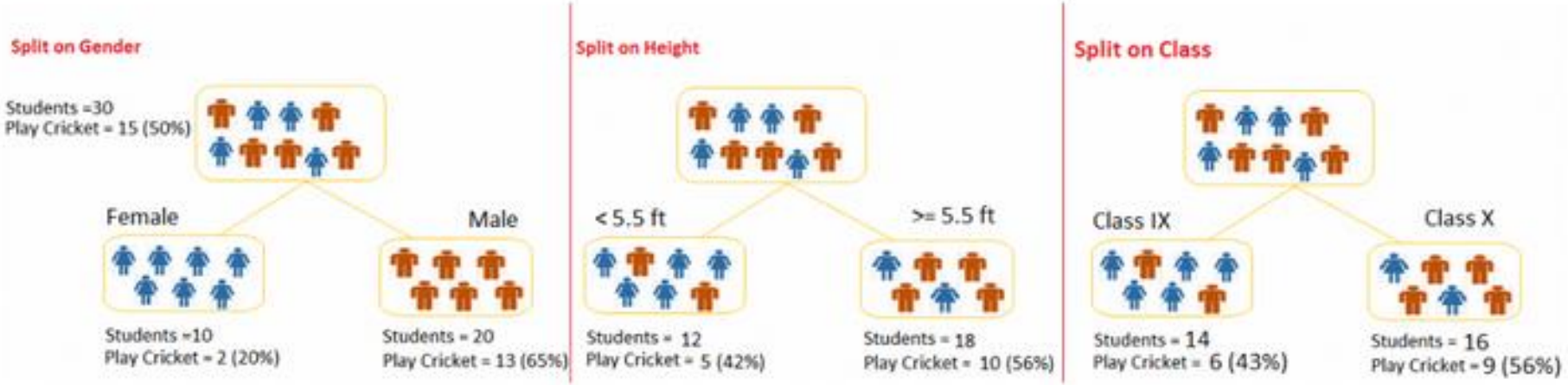



Gini value (height) = $<= 5.5 = (p)^2 + (1-p)^2$

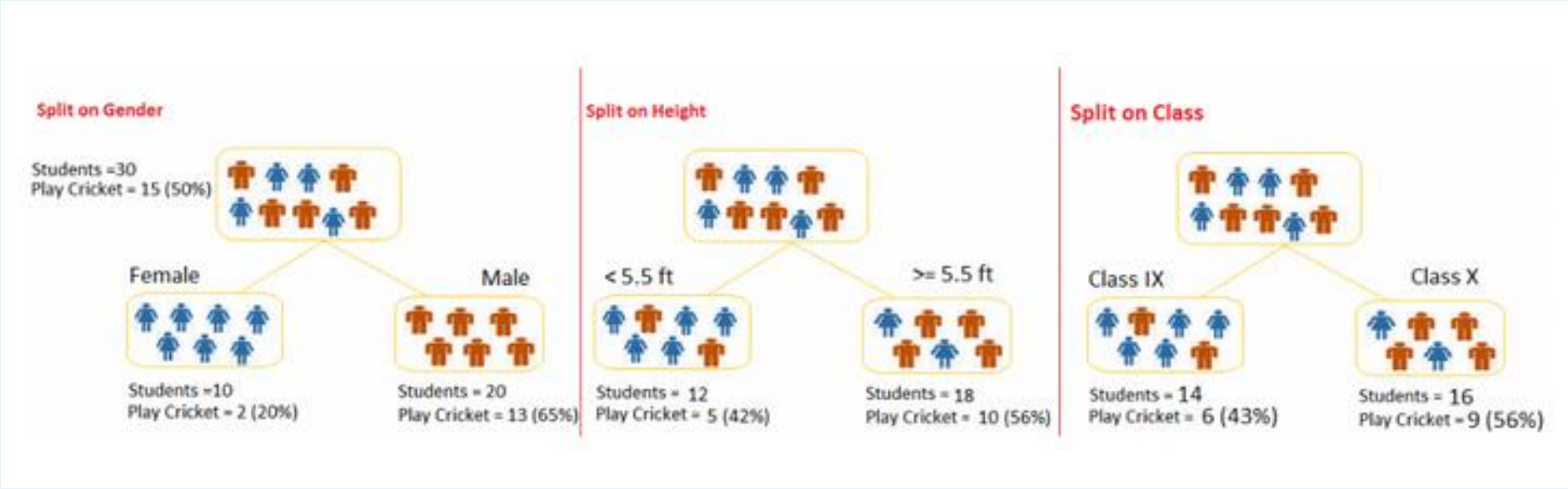
$= (0.42)^2 + (0.58)^2 = 0.5128$

$> 5.5 = (0.56)^2 + (1-0.56)^2$

$= 0.5072$

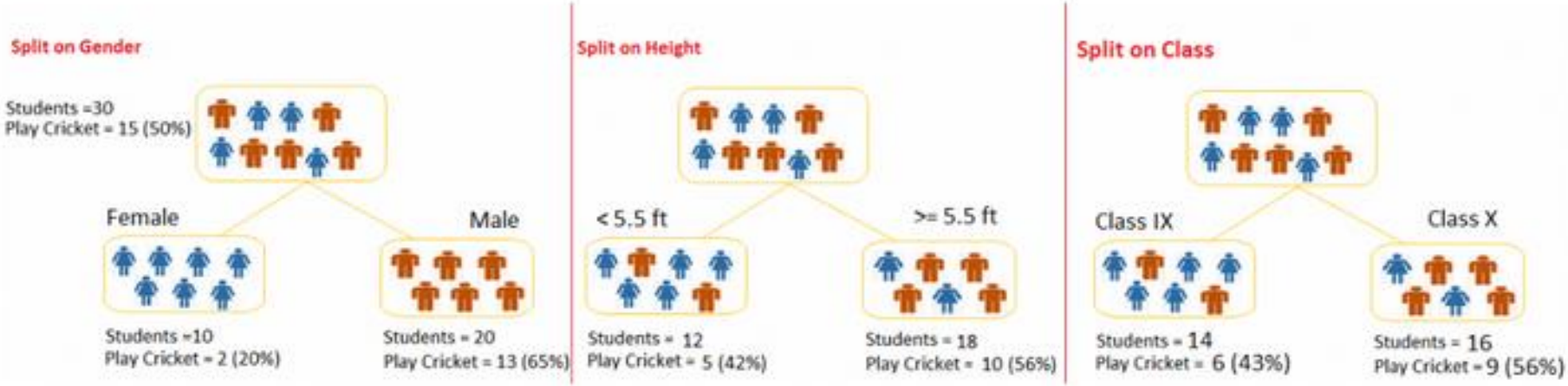



$$\text{Gini value (Height)} = \frac{12}{30} \times 0.5123 + \frac{18}{30} \times 0.5072 = 0.5094$$




Gini value (class) $\Rightarrow$ IX $\Rightarrow$ $(0.43)^2 + (0.57)^2 = 0.5098$

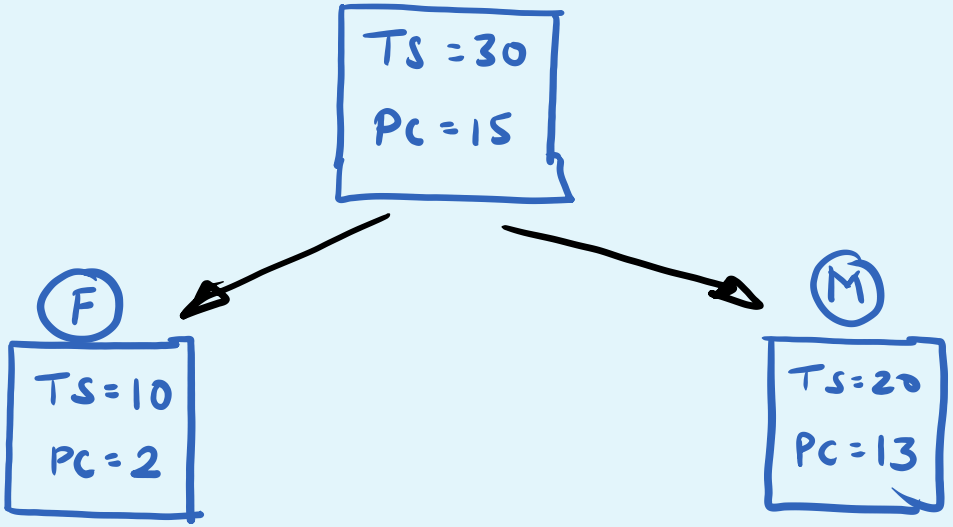
X $\Rightarrow$ $(0.56)^2 + (0.44)^2 = 0.5072$

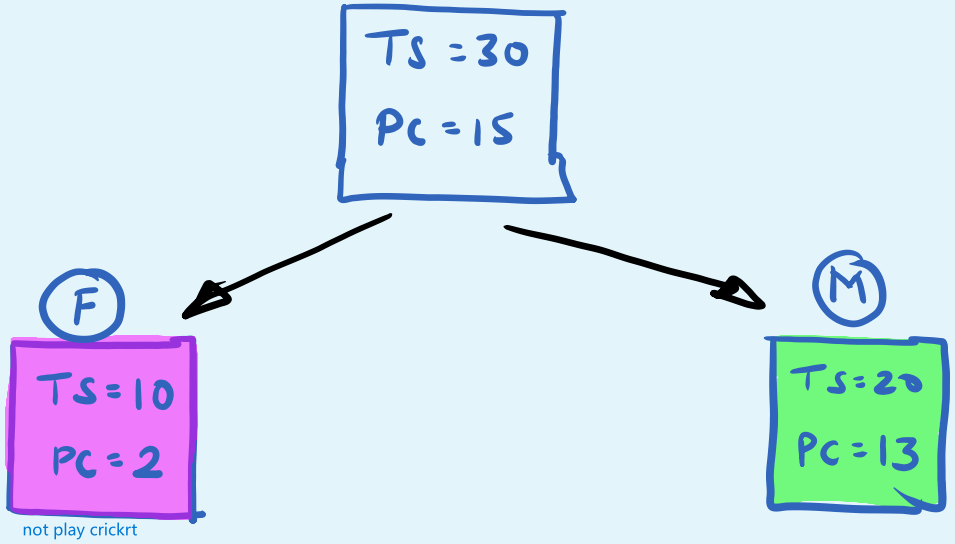



Gini value class $= \frac{14}{30} \times 0.5098 + \frac{16}{30} \times 0.5072$

$= 0.5084$

| Variable | Gini |
| --- | --- |
| Gender | 0.586 |
| Height | 0.5094 |
| class | 0.5084 |


TS = 30
PC = 15
F
TS = 10
PC = 2
M
TS = 20
PC = 13



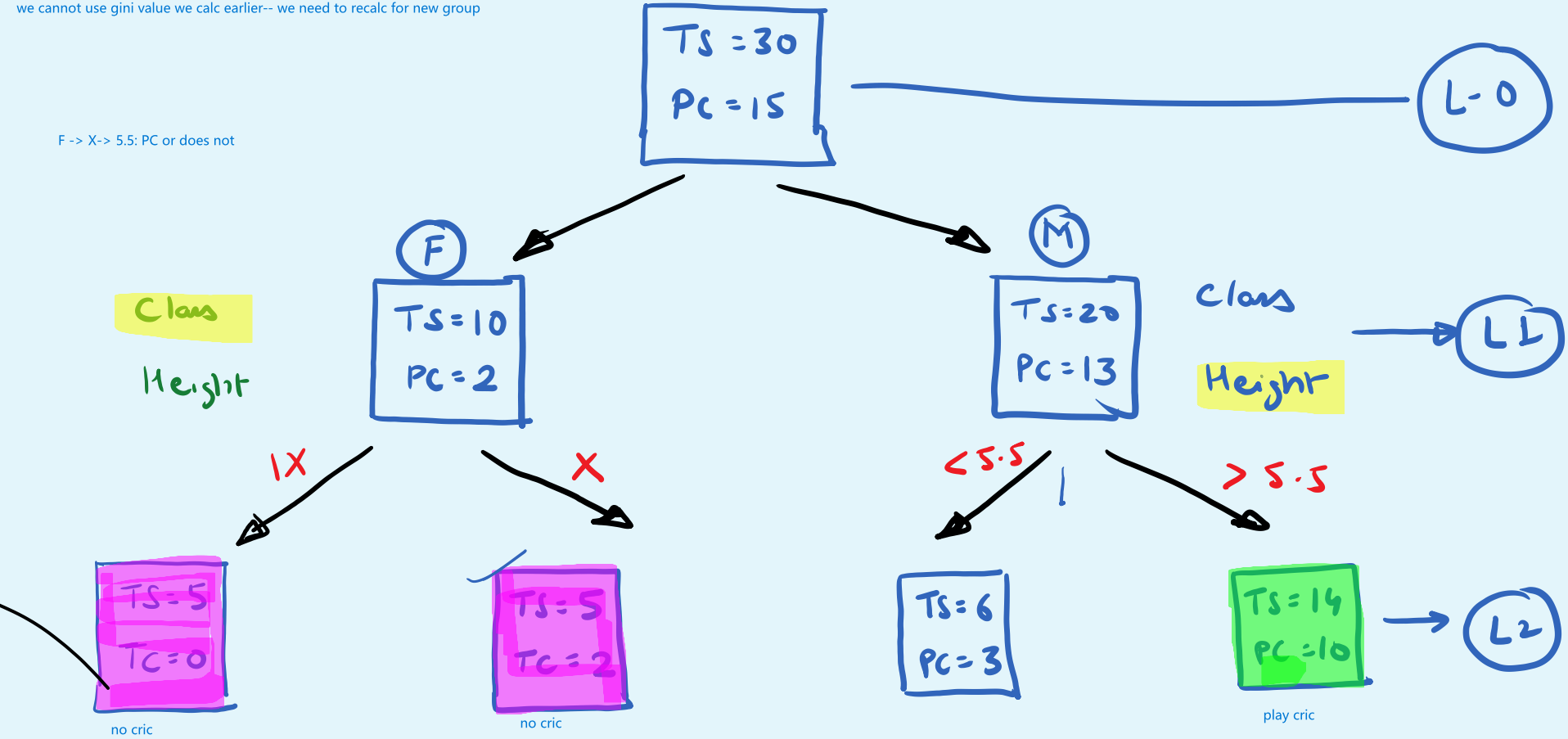

Total classification = 30

Correct classification = 21

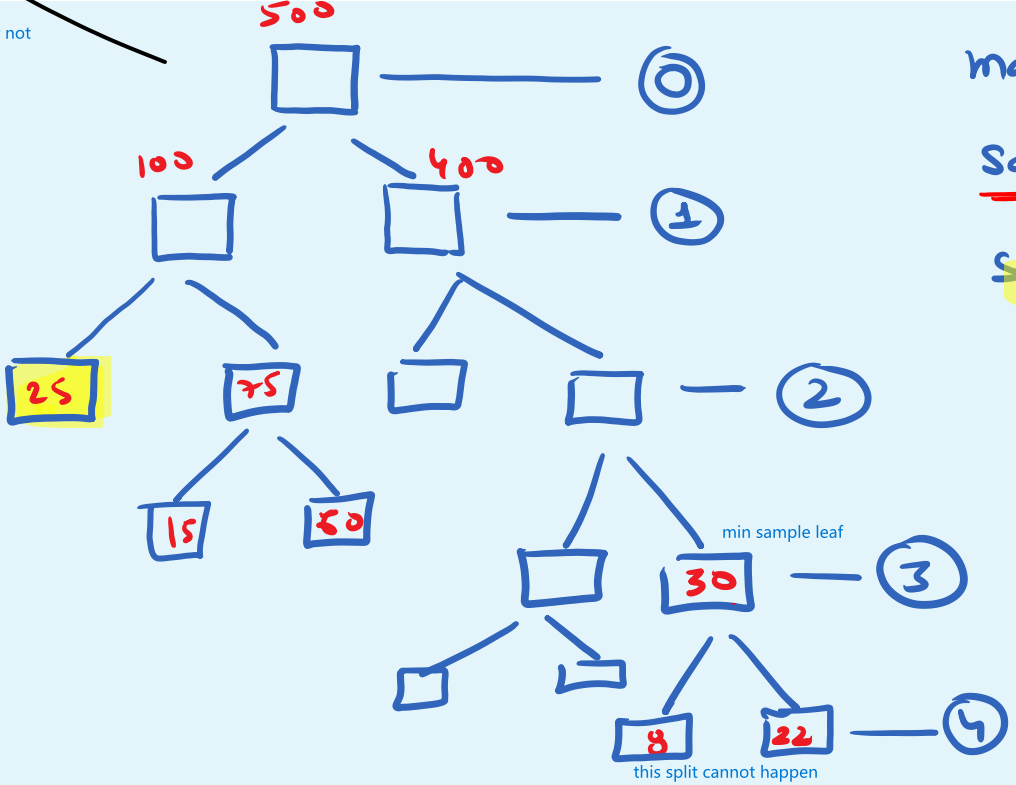
$$Acc = \frac{21}{30} = 70\%$$

we cannot use gini value we calc earlier-- we need to recalc for new group

F -> X-> 5.5: PC or does not



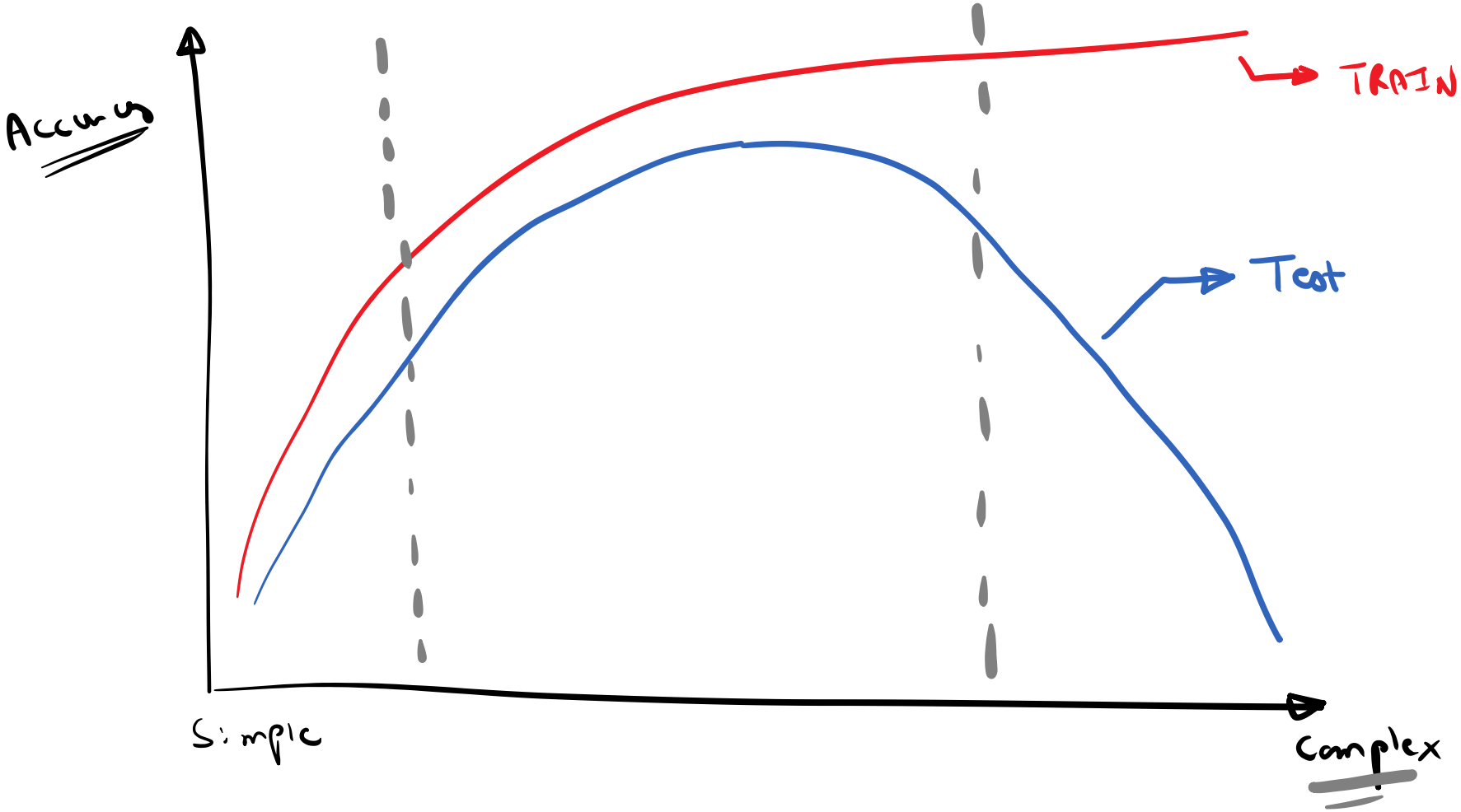
I will check every point whether my dept is 4 or not

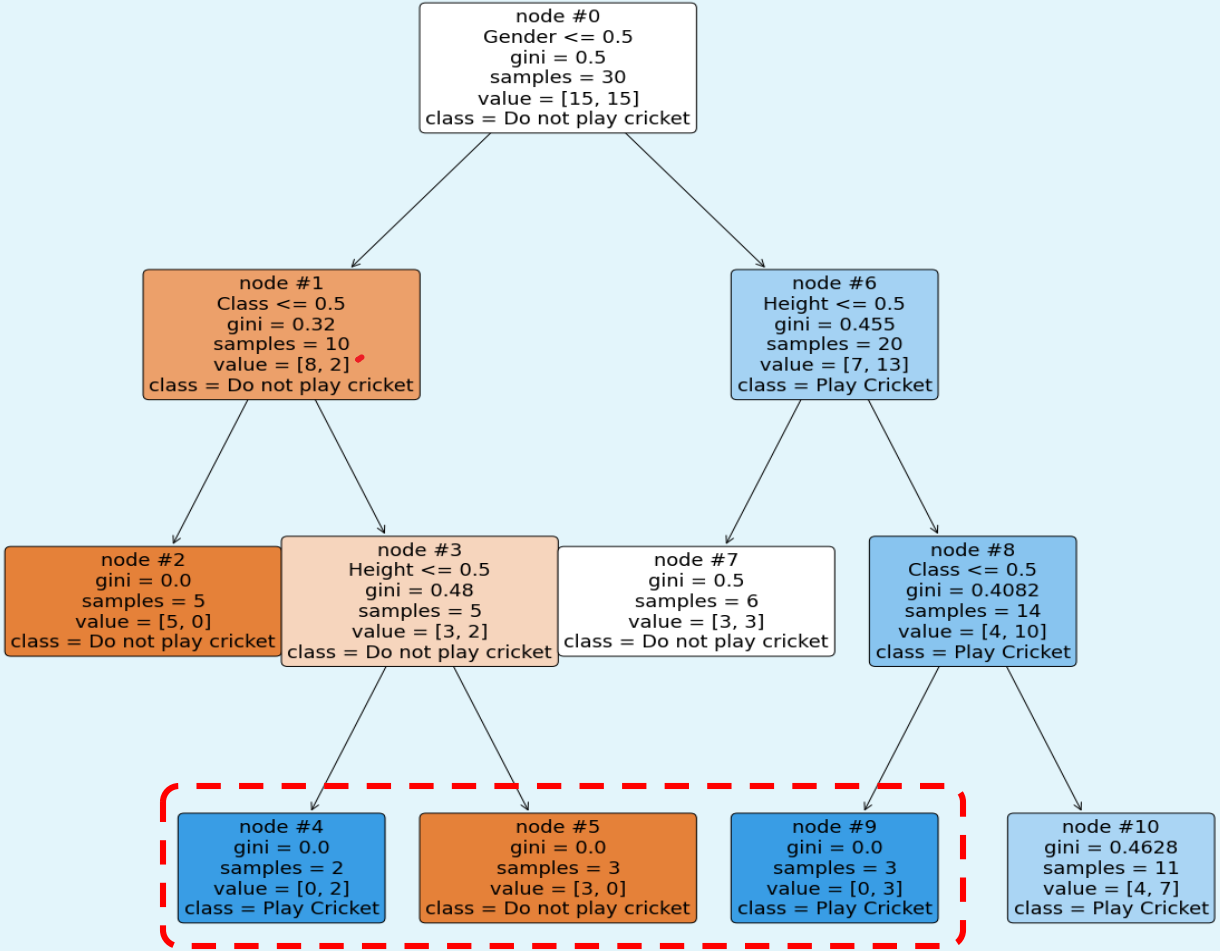


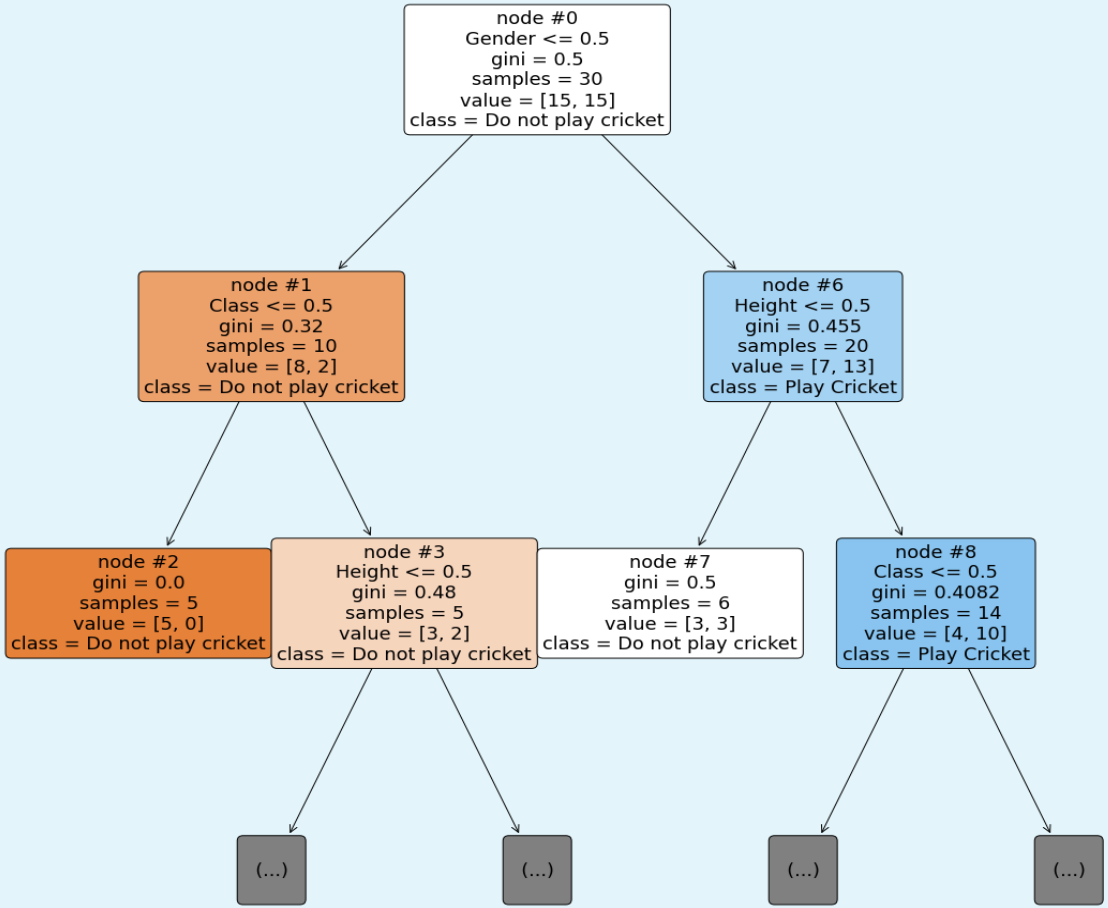

max depth = 4

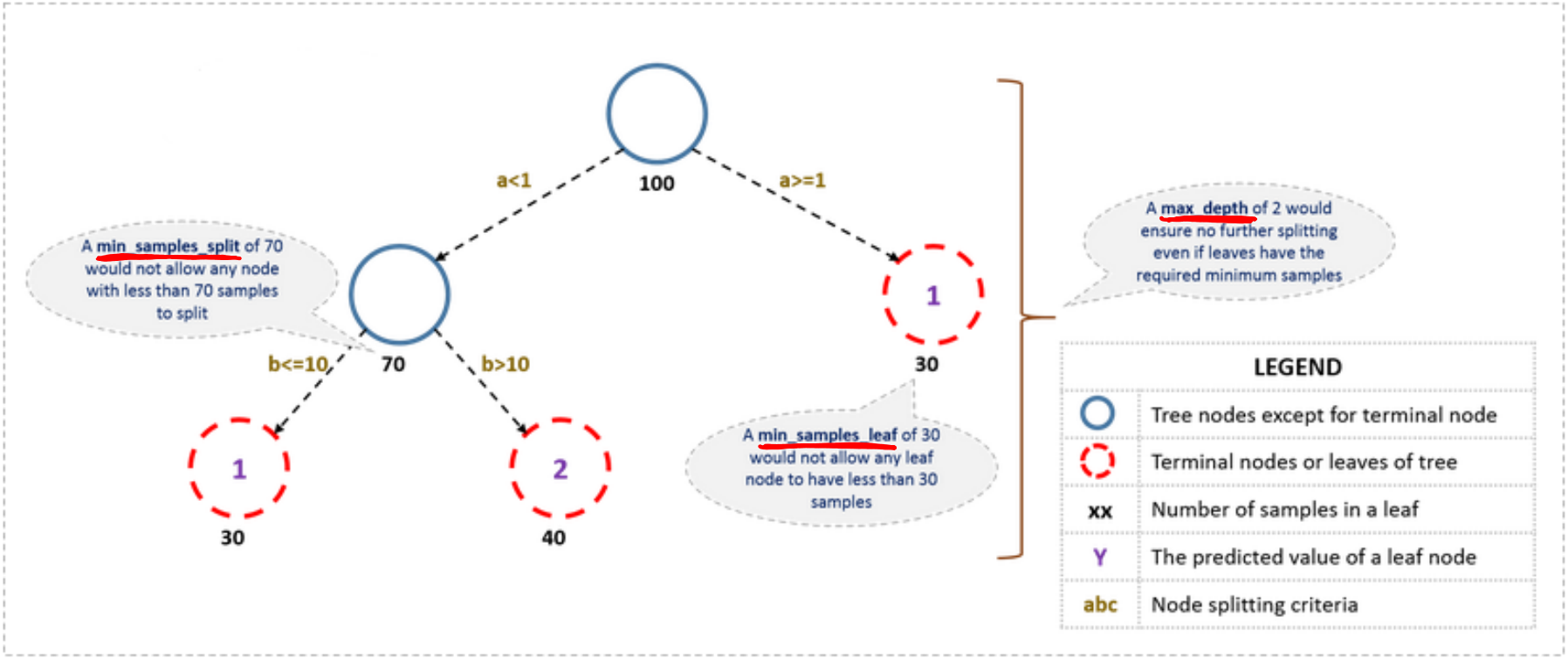
Sample leaf = 10

Sample node = 35


Accuracy
TRAIN
Test
Simple
Complex


node #0
Gender <= 0.5
gini = 0.5
samples = 30
value = [15, 15]
class = Do not play cricket
node #1
Class <= 0.5
gini = 0.32
samples = 10
value = [8, 2]
class = Do not play cricket
node #6
Height <= 0.5
gini = 0.455
samples = 20
value = [7, 13]
class = Play Cricket
node #2
gini = 0.0
samples = 5
value = [5, 0]
class = Do not play cricket
node #3
Height <= 0.5
gini = 0.48
samples = 5
value = [3, 2]
class = Do not play cricket
node #7
gini = 0.5
samples = 6
value = [3, 3]
class = Do not play cricket
node #8
Class <= 0.5
gini = 0.4082
samples = 14
value = [4, 10]
class = Play Cricket
node #4
gini = 0.0
samples = 2
value = [0, 2]
class = Play Cricket
node #5
gini = 0.0
samples = 3
value = [3, 0]
class = Do not play cricket
node #9
gini = 0.0
samples = 3
value = [0, 3]
class = Play Cricket
node #10
gini = 0.4628
samples = 11
value = [4, 7]
class = Play Cricket


node #0
Gender <= 0.5
gini = 0.5
samples = 30
value = [15, 15]
class = Do not play cricket
node #1
Class <= 0.5
gini = 0.32
samples = 10
value = [8, 2]
class = Do not play cricket
node #6
Height <= 0.5
gini = 0.455
samples = 20
value = [7, 13]
class = Play Cricket
node #2
gini = 0.0
samples = 5
value = [5, 0]
class = Do not play cricket
node #3
Height <= 0.5
gini = 0.48
samples = 5
value = [3, 2]
class = Do not play cricket
node #7
gini = 0.5
samples = 6
value = [3, 3]
class = Do not play cricket
node #8
Class <= 0.5
gini = 0.4082
samples = 14
value = [4, 10]
class = Play Cricket
(...)
(...)
(...)
(...)


100
a<1
a>=1
70
1
30
b<=10
b>10
1
30
2
40
A min_samples_split of 70 would not allow any node with less than 70 samples to split
A max_depth of 2 would ensure no further splitting even if leaves have the required minimum samples
A min_samples_leaf of 30 would not allow any leaf node to have less than 30 samples
LEGEND
Tree nodes except for terminal node
Terminal nodes or leaves of tree
xx
Number of samples in a leaf
Y
The predicted value of a leaf node
abc
Node splitting criteria